Final Project: Pick a Project

Computer Science - Week 10

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Please make sure to your UD **email** AND **name** is below. Unless your UD emails are included in this table, then you will not earn any points for this assignment when it is graded! **Every person should submit their own individual worksheet**.

Talk about each of the questions with a friend or our LLM (or both), but **make sure the words you enter below are yours** (with the obvious exceptions notwithstanding). Brainstorming is best done in full sentences, and the main goal here is brainstorming. It's worth reviewing the information in this series of pages in the Drafter documentation to gain a better understanding of the process.

Role	Name	Email
Developer (You)	Drew Keefe	dkusanyg@udel.edu
Friend (optional)		

Part 1: The Goal & Prior Implementations

1. **Identify the Goal:** What specific goal are you trying to address? Think about your passions, hobbies, and academic interests. The latter are an excellent candidate to focus on. Some majors have suggestions listed below: these were provided by faculty in those departments. Other majors could consider a focus tied to a big final assignment in one of their recent courses in their major.
    - **Biomedical Engineering** majors may consider one of the following:
        - If you have started (or completed) the Cell/Tissue Design final project in BMEG100, design an organoid, which involves combining a cell type, biomaterial, and testing method. Have the website include a description and diagram of your design concept, a testing method, a needs statement, and an impact statement relevant to their organoid. Also include a link to a paper that supports your idea and a paragraph describing how that paper supports your design concept.

        - Otherwise, create a website related to your BMEG100 Lab Report 1, *Cytoskeletal and Nuclear Staining Lab Report Guidelines*, linked here (University of Delaware account required to access).
    - **Chemistry** and **Chemical Engineering** majors may consider creating an automated chemistry assistant that takes balanced reaction equations, starting reactant masses, and a harvested product amount to determine the limiting reactant, theoretical yield, and percent yield based on principles from Chemistry 2e (OpenStax) — Chapter 4: Chemical Reactions and Stoichiometry. To avoid manually compiling atomic mass tables or writing custom string tokenizers, use the ChemPy library to parse molecular formulas into exact molecular weights and verify reaction stoichiometry against compound data from the NIH PubChem Database. The program outputs an organized breakdown of limiting reagent status, product yields, and remaining masses of unreacted species. Also use the heat of the reaction to determine if it is exothermic or endothermic, and compute the heat flow per mole of the product.
    - Majors in the **Electric & Computer Engineering** department may consider a steganography project that hides messages in images, and then retrieves those messages from uploaded images.
    - **Mechanical Engineering** majors may consider a project that somehow uses both the NumPy and SciPy Python libraries.

I want to make a website that helps people rate movies in a more precise way. Instead of picking one star rating, the site asks questions about the story, acting, visuals, sound, pacing, and enjoyment. The user scores each one from 1 to 10, and the site gives an overall rating. It also keeps a list of the movies they've rated so they can remember which ones were worth watching.

2. **Prior Implementations:** Search the internet. What existing tools or websites currently attempt to address it?

Many websites and apps already let people track and rate movies, like Letterboxd, IMDb, Trakt, and MovieLens, and some even recommend movies based on your ratings or mood. However, none of the ones I found ask a set of questions about a movie and build the overall rating from the answers, which is what my site does.

3. **Your Implementation:** Briefly describe the legitimate purpose or functionality of your site. How would you want this web application to be used?

My site helps people rate movies in a better way. After watching a movie, the user answers questions about the story, acting, visuals, sound, pacing, and how much they enjoyed it, giving each a score from 1 to 10. The site then gives an overall rating and a short message like “Must watch” or “Skip it.” I’d want people to use it to get a better understanding of rating movies they watch.

Part 2: The Input-Process-Output (IPO) Model

Think about the fundamental flow of your program.

1. **The Core Goal:** Every program has three main phases: taking input, processing data, and presenting output. In one sentence, what is the ultimate goal your program will accomplish for the user? Be specific and use CS1 concepts.

My program takes the user’s movie title and six 1-10 scores as input, uses a function to average the scores and an if statement to pick a verdict, and shows the overall rating and a list of all rated movies (stored as a list of Movie dataclass instances in State) as output.

2. Input: What specific information will the user need to provide to your program? (e.g., a city name, a numerical budget, a list of hobbies).

The user will provide: the movie title, the movie genre, six scores from 1 to 10 for story, acting, visuals, sound, pacing, and enjoyment, and a genre to filter the list of rated movies

3. Processing: At a high level, how will the computer manipulate or evaluate that data? (e.g., calculate the average, filter out options that are too expensive, compare the input against a database).

The program will turn the six scores into integers and average them to get the overall ratingl. An if statement will then compare that rating to cutoffs to pick a message like “Must watch,” “Decent,” or “Skip it.” It will also loop through the list of movies to filter them by genre and to find the average rating and the top-rated movie for the stats page.

4. Output: What will the user get back? (e.g., a recommendation, a calculated safety factor, a formatted schedule).

The user gets back an overall rating out of 10 for their movie, a short message like “Must watch” or “Skip it,” and a list of all the movies they’ve rated.

Part 3: Data Abstraction

Identify the "nouns" of your problem.

1. **Representing Reality:** Abstraction is the process of representing the real world inside a computer by ignoring unnecessary details. What are the core "things" (nouns) your program needs to track? (e.g., *If building a recipe app, the nouns are "Recipe" and "Ingredient"*).

The overall rating, a list of movies they have watched, and a stats page showing their average rating and their top-rated movie

2. **Identifying Properties:** For the most important "thing" you listed above, what 3 to 4 specific details do you actually need to solve your problem, and what details will you abstract away?
    - *Details I need:* (e.g., *Ingredient name, weight, cost*)
    - *Details I will ignore:* (e.g., *Ingredient color, brand, expiration date*)

The most important thing is a Movie. I need its title, genre, six scores, and overall rating to filter the list and give a message. I will ignore details like the director, actors, release year, and poster, because they aren’t needed to rate a movie.

Part 4: Challenges, AI Strategy, & Goals

1. **Difficulties:** What parts of this project seem the most difficult or technically complex?

I think the hardest parts will be remembering which movie the user is rating as they go from page to page, and changing the SelectBox answers from text into numbers without errors. Filtering and deleting movies from the list will also be tricky, and so will writing a test for every route.

2. **Coding Strategy:** If you will be using an LLM to help code, how will you ensure you understand the generated code and avoid "hallucinations" or "shallow learning" (copying without understanding)? If you won't be using an LLM to help code, this project might take about 4 hours a week to complete, so list each of the days and times in which you work on the project.

I plan to use an LLM for help, but I'll code in small pieces and ask it to explain anything I don't understand. I'll type the code myself, test each piece, and check it against the course rules, since an LLM can make up functions that don't exist. If I can't explain a line, I won't use it.

3. **Goals:** What is your ultimate goal for this project? (e.g., *Impress the graders with a creative UI, build something I can use in my other classes, learn Drafter deeply*).

My goal is to build a working site that meets all the requirements and that I actually understand. I also want it to be fun to use, so people will want to keep rating movies on it.

Part 5: Research & Brainstorming Conversation

Have a conversation with a Generative AI tool (like ChatGPT or Claude) OR a friend about your project idea. Do not ask for code. Instead, use the AI/friend as a sounding board to research your problem.

1. **The Initial Prompt/Question:** Ask the AI or your friend to help you brainstorm. (e.g., *"I want to build a simple web tool to help students track their study hours. What are 3 common problems students face when tracking their time, and what are 3 simple features a basic tool should have?"*). Paste your prompt below:

"I want to build a simple web tool where people rate movies by scoring the story, acting, visuals, sound, pacing, and enjoyment from 1 to 10, and the site averages the scores into an overall rating. What are 3 common problems people have when rating movies, and what are 3 simple features a basic tool like this should have? Please don't give me any code."

2. **The Response:** Briefly summarize the best ideas the AI or your friend suggested. What existing solutions or prior apps did they mention?

People struggle to rate movies fairly because an overall good feeling makes them score every category high, a 5 means different things to different people, and filling out six scores

per film gets tiring. To fix this, a simple app needs visual charts to show a movie's strengths, short hints on the score buttons so ratings stay consistent, and a searchable list to easily organize saved films.

3. **The Pushback:** You must reply to the AI/friend to narrow the scope when it's over-ambitious. Pick the most complex feature they suggested and explain why it is too complicated for an introductory programming project. Ask them for a simpler alternative. *Paste your response here:*

A radar chart or score bars sound cool, but they're too hard for an intro project. Drafter doesn't have a chart tool, and we haven't learned how to draw one. Can you suggest something simpler, like showing each score as text and pointing out the highest and lowest score?

4. **Reflection:** Based on this conversation, how has your project idea changed or become more focused?

My idea is now more focused and easier to build. I dropped the radar chart and will show each score as text and point out the highest and lowest score. I also plan to add short hints on the score choices so ratings are more fair, and a list of rated movies that can be filtered by genre. This keeps the main idea but makes it something I can actually finish.

Part 6: Resource Planning

1. What resources will you use if you get stuck?

Replace this text with your answer. If I get stuck, I'll start with the course notes. If that doesn't help, I'll use an LLM to explain error messages and ideas I don't understand, but I'll make sure I understand its answer before using it.

2. If your LLM provides code that causes a Drafter or Bakery error, or if you run into a problem during deployment, what is your step-by-step debugging plan to resolve it?

Read the error message and find the line it points to.
Remove the new code to check that it caused the problem.
Compare it to the Drafter docs and class examples.
Test small pieces and use print to see what's going on.

If I'm still stuck, ask my TA or go to office hours.

Part 7: LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

"I want to build a simple web tool where people rate movies by scoring the story, acting, visuals, sound, pacing, and enjoyment from 1 to 10, and the site averages the scores into an overall rating. What are 3 common problems people have when rating movies, and what are 3 simple features a basic tool like this should have? Please don't give me any code."

Evaluating movies across specific categories gives a far more detailed picture than a single overall score. However, category-based rating systems introduce specific user behavior challenges and design needs.

3 Common Problems When Rating Movies
The Halo Effect (Category Leakage)

Raters often struggle to separate individual elements. If a user loved the overall movie, they tend to score every category high—giving 8s or 9s across story, acting, visuals, and sound—even if the pacing was sluggish or the script was weak.

Scale Drift (Lack of Calibration)

Without clear benchmark definitions, a "5" means "average" to one person and "unwatchable" to another. Many raters naturally cluster all their scores between 7 and 10, compressing the scale and making fine comparisons between movies difficult.

Evaluation Fatigue

Scoring six distinct parameters for every single film requires effort. If the scoring form feels long or tedious, users will either stop logging movies or start picking quick, arbitrary numbers just to complete the entry.

3 Essential Features for a Basic Tool
Interactive Score Breakdown & Radar Visualization

A quick visual summary showing a movie's balance across all six attributes (like a radar

chart or color-coded score bars). This makes it easy to spot at a glance whether a film succeeded on pure enjoyment versus technical craft like visuals and sound.

Anchor Descriptions on Hover/Selection

Brief reference guides attached to scale numbers (e.g., for Story: 3 = Predictable/Flawed, 5 = Standard/Passable, 9 = Masterpiece). Clear reference points keep ratings consistent across different films over time.

Filterable Personal Movie Library

A simple database view where users can sort their saved movies by overall score, individual category performance (e.g., "Highest Visuals"), date added, or title. This turns the rating tool into a useful personal archive.

A radar chart or score bars sound cool, but they're too hard for an intro project. Drafter doesn't have a chart tool, and we haven't learned how to draw one. Can you suggest something simpler, like showing each score as text and pointing out the highest and lowest score?

That is a great way to simplify the project. Showing scores as plain text with simple high/low labels relies on basic if/else logic and basic HTML—perfect for an introductory project.

Here are 3 ways to display the ratings without any complex code or external tools:

Text Labels Next to the Scores

Display the standard score list and append a simple tag like (Best) or (Worst) next to the top and bottom numbers.

Example: Visuals: 9/10 (Highest) | Pacing: 4/10 (Lowest)

Highlighted Score Cards

Put each category score in a small text box, and use simple CSS to highlight the highest score in green and the lowest score in red.

A "Strengths & Weaknesses" Summary Box

Underneath the overall average, write two short sentences generated by your code:

"Biggest Strength: Visuals (9/10)"

"Main Weakness: Pacing (4/10)"

Part 8: Submit

In Google docs, with this document open, please click on *File* > *Download* > *Microsoft Word (.docx)*, and make sure you put that download somewhere you can find it again. Then, upload it to the Canvas assignment about selecting or brainstorming your final project.

When you're done with this assignment, review this page of the Drafter documentation to better prepare for the next step.